

1 (a) (i) Like charges repel. Since both nuclei are positively charged, they repel.

(ii) Nuclear **fusion** will occur. If v is sufficiently high, the nuclei will have sufficient kinetic energy to overcome the electrostatic repulsion.

(iii) The initial momentum of the system before collision is not zero, and according to the law of conservation of linear momentum, in the absence of external force, the total momentum of the system must be conserved. Hence the momentum of the system cannot be zero.

Note: Alternatively, we can explain that the repulsive electric forces on the charges made an action and reaction pair of the same magnitude. The decelerations of the nuclei are different in magnitude due to the different masses. With the same initial speed, the nuclei will stop at different times.

(b) Apply the law of conservation of linear momentum and let v' be their common velocity.

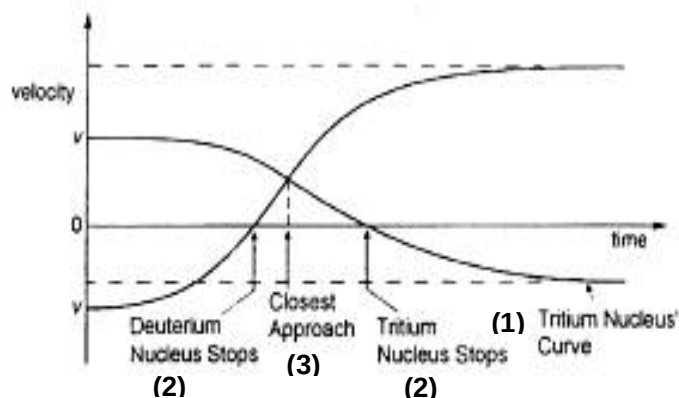
$$3mv - 2mv = (3m + 2m)v'$$

$$mv = 5mv'$$

$$v' = 0.2v$$

(c) (i) The tritium nucleus being the heavier of the two nuclei, has a smaller deceleration and experiences a smaller reversal in velocity after the collision.

The distance of closest approach occurs when both nuclei are travelling at the same speed, just before they separate.



(ii) For an elastic collision, the relative velocity of approach equals the relative velocity of separation.

$$v - (-v) = v_D - v_T$$

$$v_D - v_T = 2v \quad \text{--- (1)}$$

where v_D and v_T are the velocities of the deuterium nucleus and tritium nucleus after the collision, respectively.

From the law of conservation of linear momentum,

$$3mv - 2mv = 3mv_T + 2mv_D$$

$$3v_T + 2v_D = v \quad \text{--- (2)}$$

$$3 \times (1) + (2), \quad 5v_D = 7v$$

$$v_D = 1.4v$$

$$\text{Substitute into (1), } 1.4v - v_T = 2v$$

$$v_T = -0.6v$$

Therefore, the final speed of the deuterium nucleus is $1.4v$ while the final speed of the tritium nucleus is $0.6v$.